Artifical Intelligence Basics Course Summary

Generated by Scribe.AI

December 9, 2024

1 Key Terms

1.1 Fundamental AI Concepts

- **Inductive Reasoning**: The process of reasoning from specific observations to general conclusions.
- **Linear Regression**: A machine learning algorithm used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data.
- **Model**: A mathematical representation of a system or process, used to make predictions or understand relationships between variables.
- **Overfitting**: A phenomenon in machine learning where a model learns the training data too well, resulting in poor generalization to unseen data.
- **Decision Boundary**: A line or surface that separates different classes of data points in a classification problem.
- **0-1 Loss**: A loss function in binary classification that assigns a loss of 1 if the prediction is incorrect and 0 if it is correct.
- **Perceptron Loss**: A loss function in binary classification that is 0 if the prediction is correct and -z otherwise, where z is the margin.
- **Hinge Loss**: A loss function in binary classification that is 0 if the margin is greater than or equal to 1 and 1 - z otherwise, where z is the margin.
- **Logistic Loss**: A loss function in binary classification that is given by $log(1+e^{-z})$, where z is the margin.
- **Gradient Descent**: An iterative optimization algorithm used to find the minimum of a function by repeatedly taking steps in the direction of the negative gradient.
- **Stochastic Gradient Descent**: A variant of gradient descent that updates the model parameters using the gradient calculated from a single randomly selected data point at each iteration.

1.2 Neural Networks

- **Perceptron**: A fundamental unit in neural networks, consisting of weighted inputs, a summation node, and an activation function that produces an output based on the weighted sum of inputs and a threshold.

1.3 Search Algorithms

- **Minimax Search**: A decision-making algorithm used in game playing, where the algorithm searches the game tree to find the optimal move for the current player, assuming the opponent will also play optimally.
- **Alpha-Beta Pruning**: An optimization technique used to reduce the search space in Minimax search by eliminating branches that cannot affect the optimal decision.

1.4 Machine Learning Algorithms

- **Reinforcement Learning**: A type of machine learning where an agent learns to make decisions by interacting with an environment and receiving rewards or penalties based on its actions.
- **AlphaGo**: A computer program that achieved superhuman performance in the game of Go, combining deep learning and Monte Carlo tree search.

1.5 AI History and Philosophy

- **Turing Test**: An operational test for intelligent behavior, where a human evaluator engages in natural language conversations with both a human and a machine, attempting to distinguish them. The machine passes if the evaluator cannot reliably identify it as such.
- **Chinese Room Argument**: A thought experiment that challenges the validity of the Turing Test as a measure of true intelligence, arguing that a system can pass the test without possessing genuine understanding.

1.6 Datasets and Benchmarks

- **ImageNet Dataset**: A large database of images used for training and benchmarking computer vision models.

1.7 Linear Algebra

- **Matrix**: A rectangular array of numbers, symbols, or expressions, arranged in rows and columns.
- **Linear Transform**: A function that maps vectors from one vector space to another, satisfying $T(u+v) = T(u) + T(v)$ and $T(cu) = cT(u)$ for all vectors u, v and scalars c.
- **Matrix Multiplication**: An operation that combines two matrices to produce a third matrix, where the result's element is the dot product of a row from the first matrix and a column from the second.
- **Matrix Inverse**: For a square matrix A, its inverse A^{-1} is a matrix such that $AA^{-1} = A^{-1}A = I$, where I is the identity matrix.
- **Matrix Transposition**: An operation that swaps the rows and columns of a matrix, transforming an $m \times n$ matrix into an $n \times m$ matrix.
- **Derivative**: A measure of how a function changes with respect to a change in its input.
- **Partial Derivative**: A derivative of a function of multiple variables with respect to one variable, treating the others as constants.
- **Sigmoid Function**: A mathematical function with an S-shaped curve, often used as an activation function in neural networks; its formula is $\sigma(x) = \frac{1}{1+e^{-x}}$.

2 Problem Types

2.1 Foundational AI Concepts

- **Understanding the Nature of Intelligence**: This problem explores the definition and characteristics of intelligence, both human and artificial.
- **Defining Artificial Intelligence**: This problem involves formulating a precise definition of artificial intelligence, considering various perspectives and historical contexts.

- **Passing the Turing Test**: This problem investigates whether passing the Turing Test is sufficient evidence for a machine possessing genuine intelligence, considering counterarguments like Searle's Chinese Room.
- **Evaluating Search Tree Sizes**: This problem addresses the computational complexity of searching game trees, comparing the search space of different games and highlighting the limitations of pure search algorithms.
- **Overcoming Limitations of Search Algorithms**: This problem focuses on the need for machine learning techniques to overcome the limitations of pure search algorithms in complex games with vast search spaces.
- **Understanding the Differences Between Game Types**: This problem involves comparing and contrasting different game types (e.g., chess vs. soccer) in terms of their complexity and the challenges they pose for AI algorithms.
- **Comparing AI Tasks**: This problem explores the similarities and differences between seemingly disparate AI tasks, such as speech synthesis and game playing.
- **Understanding the AI Stack**: This problem involves understanding the different components and layers that constitute the AI stack, and how they interact with each other.

2.2 Machine Learning Algorithms

- **Understanding Perceptrons**: This problem involves understanding the structure and function of a perceptron, a fundamental building block of neural networks.
- **Understanding Reinforcement Learning**: This problem involves understanding the principles of reinforcement learning, using the example of Samuel's checkers program.
- **Linear Regression**: Finding the line that best fits a set of data points, minimizing the sum of squared distances between the points and the line.
- **Multiple Linear Regression**: Extending linear regression to include multiple independent variables to predict a dependent variable.
- **Curve Fitting**: Fitting a curve to a set of data points, potentially using polynomial functions of higher degrees.
- **Overfitting**: The phenomenon where a model fits the training data too closely, resulting in poor generalization to unseen data.
- **Binary Classification**: Classifying data points into one of two categories using a linear model and a suitable loss function.
- **Choosing a Loss Function**: Selecting an appropriate loss function (e.g., 0-1 loss, perceptron loss, hinge loss, logistic loss) for binary classification, considering differentiability and computational efficiency.
- **Gradient Descent**: An iterative optimization algorithm used to minimize the loss function by following the direction of the negative gradient.
- **Stochastic Gradient Descent**: A variant of gradient descent that updates the model parameters using a randomly selected subset of the data points at each iteration.

2.3 AI Applications and Challenges

- **Addressing Challenges in Computer Vision**: This problem examines the persistent difficulties in computer vision, despite initial optimism about its tractability.
- **Mitigating AI Safety Risks**: This problem explores the vulnerabilities of AI systems, such as their susceptibility to adversarial attacks, and the need for safety measures.
- **Addressing Ethical Concerns in AI**: This problem focuses on the ethical implications of AI, particularly the potential for bias and discrimination in AI systems.
- **Understanding the AlphaGo System**: This problem involves understanding the architecture and functionality of the AlphaGo system, which combines deep learning and search techniques.

2.4 Linear Algebra for AI

- **Representing Linear Transformations with Matrices**: Matrices are used to represent linear transformations, which map vectors from one space to another while preserving linear combinations. This is fundamental to many AI algorithms.
- **Matrix Arithmetic**: The slide covers basic matrix operations such as addition, subtraction, scalar multiplication, and multiplication, which are essential for manipulating data in AI applications.
- **Solving Systems of Linear Equations**: Systems of linear equations can be represented and solved using matrices and their inverses. This is crucial for various AI tasks involving optimization and data analysis.
- **Matrix Transposition**: Matrix transposition is a fundamental operation that swaps rows and columns, useful for data manipulation and algorithm design in AI.
- **Parallelizing Matrix Multiplication**: The slide demonstrates how matrix multiplication can be parallelized using domain decomposition, improving computational efficiency in AI applications.

2.5 Calculus for AI

- **Calculus in AI**: The lecture briefly touches upon the importance of calculus, specifically derivatives, partial derivatives, and finding maxima/minima/gradients, in AI algorithms.

3 Algorithm Solutions

3.1 Linear Algebra

- **Matrix Addition**: $A + B = C$, where $c_{ij} = a_{ij} + b_{ij}$
- **Matrix Scalar Multiplication**: $k \cdot A = B$, where $b_{ij} = k \cdot a_{ij}$
- **Matrix Multiplication**: $C_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$
- **Matrix Inverse**: $AA^{-1} = A^{-1}A = I$
- **Solving Linear Equations using Matrix Inverses**: $Ax = b \implies x = A^{-1}b$

3.2 Machine Learning Fundamentals

3.3 Artificial Neural Networks

- **Derivative of Sigmoid Function**: $\frac{d}{dx}\sigma(x) = \frac{e^{-x}}{(1+e^{-x})^2}$
- **Perceptron Algorithm**: Iteratively update the weight vector w based on misclassified data points. If $sign(w^T x_n) \neq y_n$, then $w \leftarrow w + y_n x_n$.

3.4 Game Playing AI

- **Minimax Search**: A search algorithm for two-player zero-sum games that aims to find the best move for the current player, assuming the opponent also plays optimally. It uses an evaluation function to assign scores to game states and recursively explores the game tree, alternating between maximizing the score for the current player and minimizing the score for the opponent.
- **Alpha-Beta Pruning**: A technique to reduce the search space in Minimax search by avoiding the evaluation of branches that cannot affect the optimal decision. It uses alpha (best value for the maximizing player) and beta (best value for the minimizing player) values to prune branches.